Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Scalable networks of multimodal haptic arrays for plantar sensory substitution*				
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (v), Quantum ( )				
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)			Technology Field 2 (Large Category)	
	Bio・Medical			*Bio・Medical*	
	Medium Category		Small Category	Medium Category	Small Category
	Functional Restoration / Assistive & Welfare Devices		Cognitive / Sensory Function Support Devices	Bio-electronics / Bio-Information	Remote and Home Health care Devices
Concept of Proposed Technology	Diabetic peripheral neuropathy (DPN) and age-related declines in plantar sensation are major contributors to impaired balance, altered gait, and increased fall risk in older adults. We propose to develop a wearable sensory substitution system that captures plantar pressure distributions using sensor insoles and transforms that information into high-density spatiotemporal haptic patterns delivered to a skin site with intact sensation (e.g., forearm, leg, or trunk) [1]. Our core innovation is a scalable network of skin-conformable, wirelessly coordinated multimodal haptic arrays capable of delivering vibrotactile and thermal cues with fine spatial resolution and programmable patterns. The system will integrate: (1) pressure-sensing insoles, (2) multimodal haptic "receivers," and (3) a smartphone interface that configures personalized mappings, enforces safety limits, and logs data.				

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	The proposed technology aims to provide continuous, intuitive balance and gait cues during daily living, supporting fall-risk reduction outside the clinic [1,2,3,4,5].
Rationale for Joint R&D Project	A joint project is timely because (i) DPN prevalence rises with aging and diabetes burden, and falls are a leading cause of injury and loss of independence in older adults; (ii) many existing fall-prevention interventions struggle with adherence and do not provide real-time sensory cues; and (iii) commercialization depends on capabilities best delivered by an industrial partner—manufacturable wearable packaging, robust user comfort, high-reliability power/charging, cost-down component choices, QA, and scalable system integration (insole + wearable + app) 1]. A Korean SME partner can accelerate translation by driving design-for-manufacture, reliability testing, textile/encapsulation optimization, and product engineering suitable for older users (comfort, simple UX, durability), while Georgia Tech contributes the enabling multimodal haptics R&D, mapping algorithms, and validation methodology [1,2,6,7].
Key Words	1. Diabetic peripheral neuropathy (DPN) 2. Fall prevention / balance assistance 3. Plantar pressure sensing insoles 4. Multimodal haptics (thermal + vibrotactile) 5. Wearable sensory substitution

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]		**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives		We propose to develop and validate a scalable multimodal plantar sensory substitution platform that: • Captures plantar pressure distributions in real time using wearable insoles suitable for daily use. • Encodes plantar pressure into intuitive spatiotemporal vibration + temperature patterns optimized for older adults. • Delivers these patterns via skin-conformable, wirelessly networked arrays with strong safety constraints and consistent perceptual output over time. • Demonstrates feasibility and early efficacy signals on balance and gait tasks relevant to fall-risk reduction in older adults with plantar sensory loss (including DPN).			
R&D Target Specifications		Success criteria will include quantitative targets across four domains: **1) Multimodal actuation performance (safe and perceivable in older users)** • Spatially addressable vibrotactile + thermal outputs across a multi-node array • Independent control of vibration and temperature (minimal cross-coupling) • Closed-loop thermal monitoring and safety limits appropriate for older skin physiology • Repeatable perceptual output across sessions (mitigating drift and placement variability) **2) Wearability + usability (older-adult centered design)** • Comfortable, low-profile, easy-to-don form factor (simple straps/textiles) • Wireless operation with practical charging workflow • Simple smartphone UX with preset modes (standing, walking, training) and caregiver/clinician setup support as needed **3) Sensing + mapping fidelity** • Stable plantar pressure capture during standing and walking in typical footwear • Real-time mapping latency consistent with closed-loop feedback • Personalization routines to match user sensitivity and neuropathy severity (gain, pattern resolution, preferred body site) **4) Validation outcomes (pilot level, fall-relevant endpoints)** • Demonstrated discriminability of spatial patterns (vibration, thermal, combined) in older-adult cohorts • Feasibility during functional tasks tied to fall risk (quiet standing, perturbed stance training paradigms, gait symmetry proxies, obstacle negotiation paradigms as appropriate) • User acceptance signals: comfort, perceived benefit, willingness for repeated use			

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R&D Plan		**Milestone 1 (Months 0–12): Core hardware + safety foundations (DPN-oriented constraints)** • Design and fabricate first-generation multimodal haptic nodes and array architecture • Implement closed-loop thermal control with conservative safety envelopes (skin temperature limits, fault detection) • Bench characterization (thermal response, vibrotactile control, reliability screening) • Develop first insole sensing prototype with stable signal conditioning for daily footwear use **Milestone 2 (Months 12–24): Scalable networking + personalized encoding** • Expand to multi-array wireless operation and scalable addressing • Develop pressure-to-haptics encoding algorithms tuned for older users (clear, low-cognitive-load “pattern language”) • Build smartphone setup workflow (calibration, presets, session logging) • Controlled perception studies in older adults to verify discriminability and multimodal separability **Milestone 3 (Months 24–36): Pilot feasibility validation for fall-risk reduction** • Integrate wearable packaging suitable for repeated daily use (textiles/encapsulation; sweat, cleaning, long-wear comfort) • Pilot functional assessments relevant to falls (balance and walking tasks) in older adults with plantar sensory loss / DPN-relevant cohorts • Evaluate durability, adherence drivers, and real-world usability constraints • Produce design-for-manufacture recommendations and commercialization roadmap inputs (cost targets, reliability, product segmentation)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Expected R&R of Research Institute (Georgia Tech)** • Multimodal haptic array R&D: node architecture, control electronics, firmware, safety algorithms • Pressure-to-haptics encoding algorithms optimized for older adults (personalization, low cognitive load) • Validation design: perceptual tests + fall-relevant functional assessments • Data analysis and performance benchmarking; translational study planning inputs
Expected R&R of Korean SME Partner	**Expected R&R of Korean SME Partner** • Design-for-manufacture engineering: miniaturization, component selection, assembly processes • Wearable packaging for older adults: comfort, skin safety, repeatable placement, easy don/doff • System integration: charging/power, enclosures, manufacturable insole form factor options • Reliability/QA: accelerated life testing, failure modes, supply chain and cost-down plans • Product pathway inputs: fall-prevention positioning, distribution channels, early regulatory/reimbursement strategy if applicable

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes and Impact** **Technical deliverables** • Integrated prototype system: sensor insoles + scalable multimodal haptic arrays + smartphone configuration/logging • Verified multimodal performance (independence, safety limits, repeatability) under conditions relevant to older-adult daily wear • Demonstrated feasibility of real-time plantar-to-skin substitution with pilot balance/gait outcomes **Clinical and societal impact** • New pathway to reduce falls by restoring real-time plantar "awareness" through intact skin channels • Supports independence and mobility confidence in older adults with DPN and age-related sensory decline • Potential to complement existing care (foot care, exercise programs, PT) with a real-time assistive layer **Commercialization relevance** • Partner-driven manufacturability + reliability plan (materials, textiles, robustness, cost targets) • Positioned for next-stage scale-up, expanded trials, and product definition (home use vs clinic program vs wellness)

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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